

Fixed Effects Illustration: Minimum Wage and Unemployment

ECON 4999B Course Material

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1 Data

We use a balanced panel of 26 regions (A–Z) and 27 years (2000–2026), for $26 \times 27 = 702$ observations. The outcome variable is `unemployment_rate` (percent) and the regressor of interest is `min_wage` (dollars).

2 Specifications

We estimate

$$\text{unemployment_rate}_{it} = \alpha + \beta \text{min_wage}_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

where μ_i denotes region fixed effects and λ_t denotes year fixed effects. The four columns differ by which fixed effects are included:

- (1) pooled OLS (no fixed effects);
- (2) year fixed effects only;
- (3) region fixed effects only;
- (4) region and year fixed effects (two-way FE).

All models use heteroskedasticity-robust standard errors (HC1).

Table 1: Effect of minimum wage on unemployment under alternative specifications

	(1)	(2)	(3)	(4)
<i>Dependent variable: unemployment rate (%)</i>				
Min wage	-0.371*** (0.023)	-0.530*** (0.017)	0.039 (0.030)	0.174** (0.081)
Fixed effects	None	Year	Region	Region, Year
Observations			702	
R^2	0.237	0.702	0.588	0.941
Adjusted R^2	0.236	0.690	0.572	0.936

Notes: HC1 robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 Results

4 Discussion

Without fixed effects, the estimated minimum-wage coefficient reflects both within-region variation and persistent cross-region differences. Regions with historically higher minimum wages also differ in other ways (industry mix, demographics, labor-market institutions), so the pooled estimate can be misleading.

Adding region fixed effects (column 3) uses only within-region changes in the minimum wage, removing time-invariant regional confounders. Adding year fixed effects (column 2) removes common macro shocks shared by all regions in a given year. Two-way fixed effects (column 4) combine both: identification comes from region-specific deviations from national trends.

In this synthetic dataset, the pooled estimate is negative because high-wage regions have lower baseline unemployment for reasons unrelated to minimum-wage changes. Once region fixed effects are included, the coefficient turns positive, consistent with the within-region relationship built into the data-generating process.